

Figure 1: A typical VANET scenario

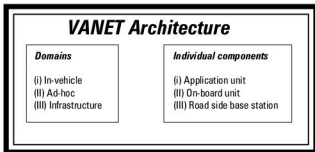


Figure 2: VANET architecture

and Simulation of Urban Mobility (SUMO) to simulate the mobility component.

For the identification of a traffic simulator, the following parameters are considered:

- Accurate and realistic topological maps: geographical maps obtained from TIGER, GDF, GPSTrack, Google Maps or ESRI.
- The database.
- Obstacles: The hurdles to a vehicle's mobility and communication.
- The smoothness of deceleration and acceleration.
- Human driving patterns.
- Intersection management, which is based on some stochastic model.
- Time patterns: The consideration of peak hours and time constraints.
- External influences, for example, road maintenance, traffic conditions and accidents.

Considering the aspects mentioned here, SUMO can be said to have most of the desired features for mobility simulation with visualisation support. SUMO helps in generating the first phase of VANET simulation, wherein we generate the road map and the vehicles' flow, which is known as a mobility pattern. The files needed to generate a mobility pattern using SUMO are:

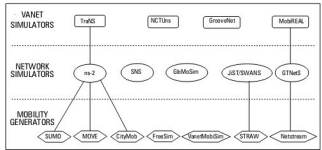


Figure 3: VANET simulators

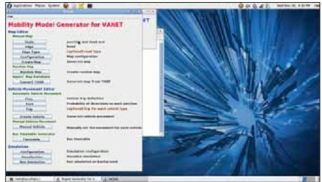


Figure 4: MOVE main screen for mobility pattern generation

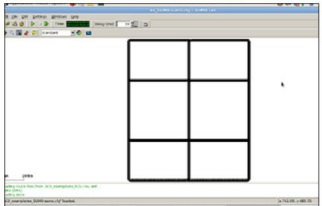


Figure 5: Sample SUMO output

1. <Filename>.nod.xml: Defines nodes
2. <Filename>.edge.xml: Defines edges
3. <Filename>.netc.cfg
4. <Filename>.net.xml: Generated using the net-convert command, from the above two files
5. <Filename>.ow.xml: Defines traffic flow definition
6. <Filename>.rou.xml: Defines vehicle routes
7. <Filename>.sumo.cfg: Simulation configuration

Generating so many files for a large number of vehicle nodes is cumbersome, so instead of typing the file contents in the specific formats, we can use another open source project called Mobility generation for Vehicular Environment (MOVE). This provides a GUI to generate SUMO outputs—also, the defined road mobility is converted to NS2-readable form (.tcl). NS2 then can be used to simulate the networking-

(Continued on Page 89....)